PORTOFLIO PROJECT

Comparative analysis of NIFTY 50 Index and its constituents from 2015 to 2025 with Pre COVID and Post COVID impact.

Business Questions and Insights.

Q1. *Which regime had better performance, was it Pre-COVID or Post-COVID?*

	market_regime text	annualized_return numeric	volatility numeric	volume numeric	risk_adjusted_return numeric
1	Post-COVID	17.73	0.0114	369194.32	15.53
2	Pre-COVID	6.94	0.0087	266300.48	7.98

Returns Nearly Tripled Post-COVID

- Pre: ~7%
- Post: ~18

Risk Increased

- Volatility ↑ from 0.0087 → 0.0114
- But returns ↑ much more

(Heightened investor participation despite increase in volatility)

Risk-Adjusted Performance Doubled

- 7.98 → 15.53

(Investor was rewarded for the risk taken)

Participation Increased

- Volume ↑ ~38%

(Volume of trades increased showcasing higher participation of investors which in term supports increase in trade capital)

Post-COVID markets delivered significantly higher and more efficient returns, supported by increased participation, despite higher volatility

Q2: Which stocks delivered the best risk-adjusted returns and in which regime?

(**"Best bang for buck" considering risk + time**)

	market_regime text	ticker text	trading_days bigint	risk_adjusted_return numeric
1	Pre-COVID	BAJFINANCE....	593	26.10
2	Pre-COVID	HINDUNILVR....	1270	20.88
3	Pre-COVID	HDFCBANK.NS	1270	20.86
4	Pre-COVID	BAJAJFINSV....	470	20.72
5	Pre-COVID	RELIANCE.NS	1270	18.25
6	Pre-COVID	KOTAKBANK....	1270	16.52
7	Pre-COVID	ASIANPAINT....	1270	15.44
8	Pre-COVID	BPCL.NS	1270	12.32
9	Pre-COVID	TITAN.NS	470	12.07
10	Pre-COVID	MARUTI.NS	1270	11.35

During the Pre-Covid period BAJAJFINANCE, HINDUUNILVR, HDFCBANK, BAJAJFINSV, RELIANCE, KOTAKBANK, ASIANPAINT, BPCL, TITAN AND MARUTI having a trading period of greater than 252 days are the top 10 stocks which delivered the best returns per unit risk.

	market_regime text	ticker text	trading_days bigint	risk_adjusted_return numeric
1	Post-COVID	SHRIRAMFIN....	436	29.41
2	Post-COVID	M&M.NS	1446	25.55
3	Post-COVID	SUNPHARMA....	1446	23.00
4	Post-COVID	BEL.NS	312	19.50
5	Post-COVID	EICHERMOT.NS	1446	19.49
6	Post-COVID	BHARTIARTL....	1446	19.43
7	Post-COVID	TATASTEEL.NS	1446	19.20
8	Post-COVID	HINDALCO.NS	1446	19.10
9	Post-COVID	JSWSTEEL.NS	1446	18.97
10	Post-COVID	GRASIM.NS	1446	18.74

During the Post-Covid period, SHRIRAMFIN, M&M, SUNPHARMA, BEL, EICHERMOT, BHARTIARTL, TATASTEEL, HINDALCO, JSWSTEEL AND GRASIM, having a trading period of greater than 252 days are the top 10 stocks which delivered the best returns per unit risk.

Risk-adjusted leadership shifted from defensive quality stocks pre-COVID to cyclical and growth sectors post-COVID, reflecting a structural change in market behaviour.

Q3: Was it riskier to trade Pre-COVID or Post-COVID?

(Volatility + Liquidity Perspective)

Index level volatility and volume

	market_regime text	avg_volatility numeric	avg_volume numeric
1	Post-COVID	0.0114	369194.32
2	Pre-COVID	0.0087	266300.48

- Volatility ↑ ~30%
- Volume ↑ ~38%

At the index level, post-COVID markets became significantly more liquid but also more volatile signifying more activity and more risk.

Stock level volatility, volume dispersion and volume

	market_regime text	avg_volatility numeric	volume_dispersion numeric	avg_volume numeric
1	Pre-COVID	0.0191	0.0047	8068290.21
2	Post-COVID	0.0204	0.0075	10743105.64

- Volatility ↑ ~30%
- Average volatility ↑ ~7%
- Dispersion (0.0047 → 0.0075)

Not only did risk increase at stock level but there was increase in overall dispersion as well, signifying uneven risk across different stocks.

Post-COVID markets became more liquid but also more volatile at the index levels. At the stock level, volatility dispersion increased significantly, indicating uneven risk distribution across constituents. This post-COVID markets more attractive for active traders but riskier for conservative investors, signifying that markets were riskier but also smoother to trade.

Q4: Which stocks outperformed/underperformed the NIFTY 50 Index?

Distribution of Performance

	performance_flag text	stock_count bigint
1	Market Performer	5
2	Outperformer	36
3	Underperformer	33

Only about half of NIFTY 50 constituents managed to beat the index, while the other half underperformed, confirming that consistent outperformance is difficult.

"Moderate Return, Long Exposure"

	ticker text	trading_days bigint	stock_annualized_return numeric	index_annualized_return numeric	excess_annualized_return numeric
1	SHRIRAMFIN....	436	52.00	11.00	41.00
2	BEL.NS	312	31.00	11.00	20.00
3	BAJFINANCE....	2039	23.00	11.00	12.00
4	TITAN.NS	1916	21.00	11.00	10.00
5	RELIANCE.NS	2716	21.00	11.00	10.00
6	LTIM.NS	298	20.00	11.00	9.00
7	M&M.NS	2716	19.00	11.00	8.00
8	SBILIFE.NS	1303	19.00	11.00	8.00
9	BHARTIARTL....	2716	19.00	11.00	8.00
10	BAJAJFINSV....	1916	19.00	11.00	8.00

SHRIRAMFIN, BAJAJFINANCE, TITAN, RELIANCE, M&M, BHARTIARTL, BAJAFINSV, SBILIFE ADANIPORTS, HINDALCO are the top 10 stocks which outperformed the Index having a trading period greater than 252 days.

“High Return, Short Exposure companies”

	ticker text	trading_days bigint	stock_annualized_return numeric	index_annualized_return numeric	excess_annualized_return numeric
1	ETERNAL....	186	53.00	11.00	42.00
2	JIOFIN.NS	186	38.00	11.00	27.00

ETERNAL, JIOFIN, BEL and LTIM are the only to stocks which outperformed despite having a short trading period of less than 252 days.

“Short Exposure, Underperformance”

	ticker text	trading_days bigint	stock_annualized_return numeric	index_annualized_return numeric	excess_annualized_return numeric
1	NMDC....	183	-42.00	11.00	-53.00

NMDC is the only stock which underperformed having a trading period of less than 252 days.

“Long Exposure, Underperformance”

	ticker text	trading_days bigint	stock_annualized_return numeric	index_annualized_return numeric	excess_annualized_return numeric
1	PNB.NS	307	-52.00	11.00	-63.00
2	TRENT.NS	312	-38.00	11.00	-49.00
3	SAMMAANCAP....	613	-29.00	11.00	-40.00
4	IDEA.NS	553	-24.00	11.00	-35.00
5	HINDPETRO.NS	368	-24.00	11.00	-35.00
6	BHEL.NS	553	-21.00	11.00	-32.00
7	YESBANK.NS	1225	-20.00	11.00	-31.00
8	VEDL.NS	783	-17.00	11.00	-28.00
9	BANKBARODA....	677	-15.00	11.00	-26.00
10	ADANIENT.NS	803	-10.00	11.00	-22.00

PNB, TRENT, SAMMANCAP, HINDPETRO, IDEA, BHEL, YESBANK, VEDL, BANKBARODA, ADANIENT are top 10 stocks which underperformed having a trading period of more than 252 days.

While several stocks generated high excess returns over short periods, only a few large-cap leaders consistently outperformed the index over long membership periods, highlighting the difficulty of sustained alpha generation.

Q5. How Severe were Downside Risks?

(Maximum Drawdowns of Stocks vs Index, Pre vs Post COVID)

Index level Maximum Drawdown

	market_regime text	max_drawdown numeric
1	Post-COVID	-33.00
2	Pre-COVID	-23.00

Post COVID market corrections were deeper even though returns were higher. This suggests returns also come with a downside risk.

Top 10 stocks with their respective max drawdown - Pre COVID

	market_regime text	ticker text	max_drawdown numeric
1	Pre-COVID	YESBANK.NS	-92.00
2	Pre-COVID	SAMMAANCAP....	-70.00
3	Pre-COVID	BHEL.NS	-69.00
4	Pre-COVID	SUNPHARMA.NS	-68.00
5	Pre-COVID	PNB.NS	-67.00
6	Pre-COVID	IDEA.NS	-66.00
7	Pre-COVID	LUPIN.NS	-65.00
8	Pre-COVID	HINDPETRO.NS	-64.00
9	Pre-COVID	ZEEL.NS	-62.00
10	Pre-COVID	HINDALCO.NS	-61.00

Top 10 stocks with their respective max drawdown - Post COVID

	market_regime text	ticker text	max_drawdown numeric
1	Post-COVID	INDUSINDBK.NS	-73.00
2	Post-COVID	ADANIENT.NS	-71.00
3	Post-COVID	BAJFINANCE....	-59.00
4	Post-COVID	AXISBANK.NS	-56.00
5	Post-COVID	BAJAJFINSV.NS	-54.00
6	Post-COVID	ADANIPORTS....	-52.00
7	Post-COVID	UPL.NS	-51.00
8	Post-COVID	TRENT.NS	-51.00
9	Post-COVID	WIPRO.NS	-50.00
10	Post-COVID	HINDALCO.NS	-50.00

While post-COVID markets delivered higher returns, they also experienced deeper index-level drawdowns. At the stock level, extreme losses occurred in both regimes, highlighting significant risk in individual stocks, reinforcing the idea of portfolio diversification and risk management.

Summary and Findings

- The goal was to understand how market regimes change by comparing pre-COVID (2015–2020) and post-COVID (2020–2025) periods using metrics like returns, volatility, liquidity, drawdowns, and risk-adjusted performance.
- I found that post-COVID markets delivered nearly three times higher returns, but volatility also increased by around 30%. Interestingly, risk-adjusted returns improved significantly, meaning investors were actually better compensated for the additional risk.
- Another key insight was the shift in market leadership — pre-COVID performance was driven by defensive, high-quality large caps like FMCG and private banks, whereas post-COVID leadership moved to cyclical and growth sectors such as metals, capital goods, and telecom. This showed a structural change in investor behaviour.
- I also discovered that only about half of the NIFTY 50 stocks consistently outperformed the index, reinforcing how difficult it is to generate sustained alpha in large-cap markets.
- Overall, the project helps us understand how macro events reshape risk, liquidity, and sector leadership, and it strengthened my ability to analyse markets not just from a returns perspective, but from a risk-reward and decision-making standpoint using Python, pandas, SQL and financial metrics.

Overall Insights

Metric	**Pre-COVID**	**Post-COVID**
Returns	Moderate	Very High
Volatility	Lower	Higher
Liquidity	Lower	Much Higher
Risk-Adjusted Returns	Good	Better
Market Leadership	Defensive Stocks	Cyclical/Growth Stocks
Drawdowns	Shallower	Deeper
Investor Participation	Moderate	Extremely High

Financial Metrics Calculated in the Project.

Metric	**Purpose**
Daily Returns	Measure day-to-day price movement
Average Returns	Evaluate overall profitability
Volatility (Standard Deviation)	Measure market risk/fluctuation
Risk-Adjusted Return	Return earned per unit of risk
Trading Volume	Measure market participation/liquidity
Volume Dispersion	Understand uneven participation across stocks
Volatility Dispersion	Measure uneven risk across constituents
Maximum Drawdown	Measure worst peak-to-trough decline
Excess Return vs NIFTY	Identify outperformers/underperformers
Trading Exposure (Days >252)	Identify sustained long-term performers

Metric	Purpose
Index vs Stock Comparison	Benchmark constituent performance
Pre vs Post Regime Comparison	Measure structural market changes